

Family Name

Given Name

Student No.

Signature

UNSW SYDNEY

School of Electrical Engineering & Telecommunications

FINAL EXAMINATION

Semester 1, 2017

ELEC1111

Electrical and Telecommunications Engineering

TIME ALLOWED:	2 hours
TOTAL MARKS:	100
TOTAL NUMBER OF QUESTIONS:	5

THIS EXAM CONTRIBUTES 50% TO THE TOTAL COURSE ASSESSMENT

Reading Time: 10 minutes.

This paper contains 6 pages.

Candidates must **ATTEMPT ALL** questions.

Answer each question in a **separate answer booklet**.

Marks for each question are indicated beside the question.

This paper **MAY NOT** be retained by the candidate.

Students must achieve a **minimum of 40 marks** to pass the course.

Print your name, student ID and question number on the front page of each answer book.

Authorised examination materials:

Candidates should use their own UNSW-approved electronic calculators.

This is a closed book examination.

Assumptions made in answering the questions should be stated explicitly.

All answers must be written in ink. Except where they are expressly required, pencils **may only be used** for drawing, sketching or graphical work.

QUESTION 1 [20 marks]

- (i) [10 marks] In the circuit of Figure 1,
- [5 marks] Calculate the voltage V_x .
 - [3 marks] Calculate the power dissipated in the $4\ \Omega$ resistor.
 - [2 marks] Does the dependent voltage source in this circuit supply or absorb power? Explain your answer.

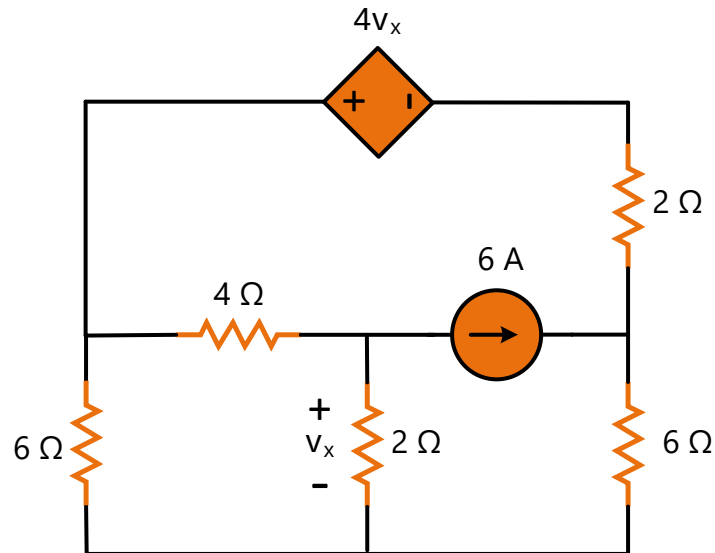


Figure 1

- (ii) [10 marks] For the circuit of Figure 2
- [4 marks] Calculate the Thevenin voltage for terminals a-b
 - [4 marks] Find the Thevenin resistance R_{th} for terminals a-b and draw the equivalent Thevenin circuit.
 - [2 marks] What is the maximum power that can be delivered to a load from this circuit?

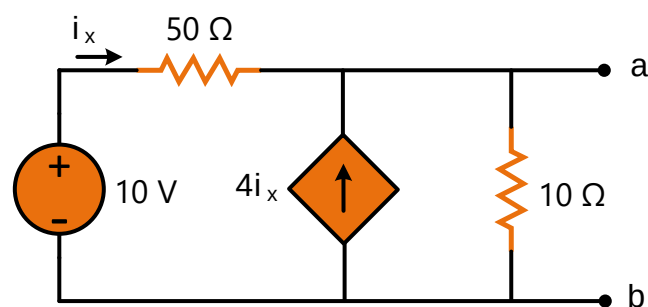


Figure 2

QUESTION 2 [20 marks]

(i) [8 marks] Operational Amplifiers

- a. [5 marks] Demonstrate, deriving the required equations, that the gain of a *non-inverting amplifier* circuit is given by

$$\frac{V_{out}}{V_{in}} = 1 + \frac{R_f}{R_{in}}$$

where R_f and R_{in} are the feedback and input resistances, respectively.

- b. [1 mark] From the following set of resistors select R_f and R_{in} to achieve a gain of 12.

1 Ω	5 Ω	8 Ω	10 Ω	20 Ω	50 Ω
60 Ω	80 Ω	100 Ω	110 Ω	120 Ω	150 Ω
100 Ω	220 Ω	240 Ω	330 Ω	470 Ω	600 Ω

- c. [2 marks] What resistors from the ones above should be used if you would like to construct an *inverting amplifier* where $\frac{V_{out}}{V_{in}} = -12$.

- (ii) [12 marks] Find the output voltage v_{out} of the following Operational Amplifier circuit. Plot the voltages v_{in} and v_{out} in the same phasor diagram.

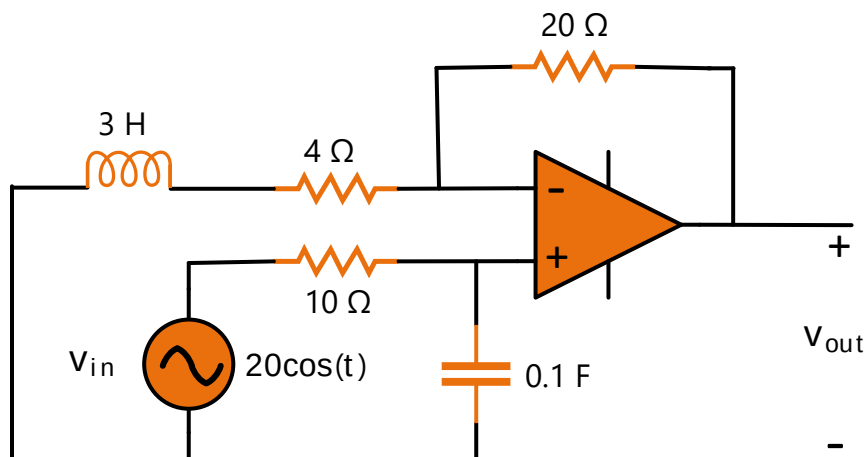


Figure 3

QUESTION 3 [20 marks]

- (i) [10 marks] Find the voltage $v_x(t)$ in the circuit of Figure 4.

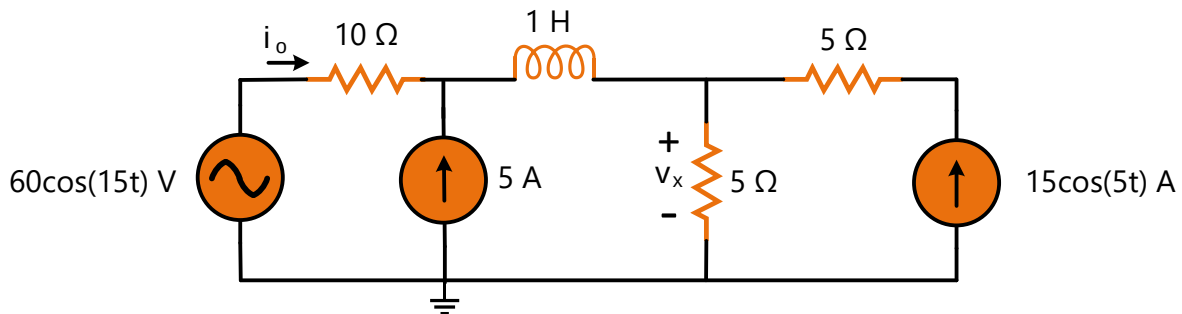


Figure 4

- (ii) [10 marks] In the following circuit of Figure 5,
- [2 marks] Find the energy stored in the inductor under steady-state when the switch is in the open position.
 - [4 marks] If the switch has been in the open position for a long time and closes at $t=0$, derive an analytical expression for the current i_L through the inductor for $t>0$.
 - [2 marks] Plot the current through the inductor as a function of time.
 - [2 marks] Derive an analytical expression for the voltages across resistors R_1 and R_2 (v_{R_1} and v_{R_2}) as a function of time for $t>0$.

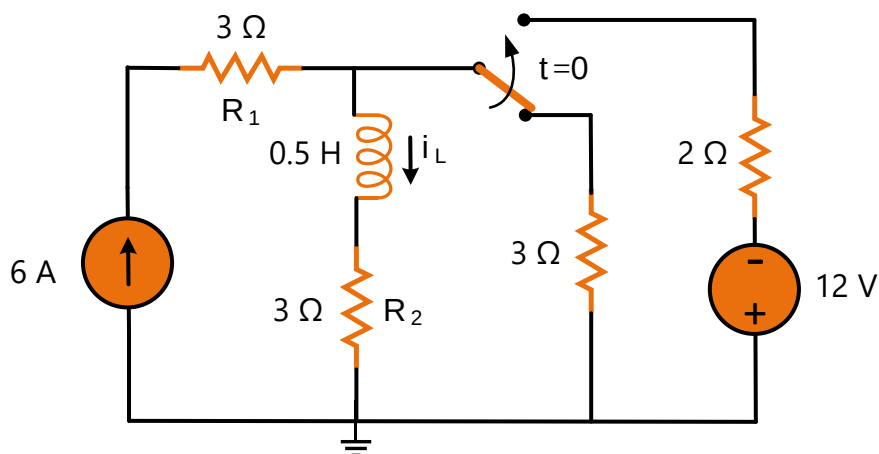


Figure 5

QUESTION 4 [20 marks]

- (i) **[8 marks]** Calculate the equivalent impedance of the circuit of Figure 6 as seen by:
- (2 marks)** a *dc* voltage source
 - (6 marks)** an *ac* voltage source with a frequency of $\omega = 2$ rad/sec. (Provide your answer in rectangular or polar form)

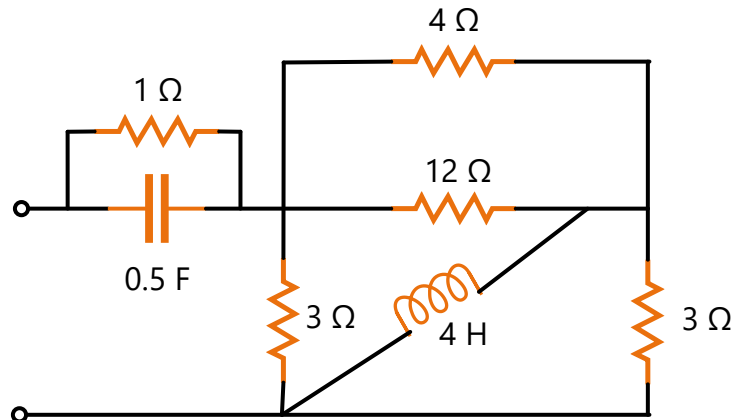


Figure 6

- (ii) **[10 marks]** For the circuit shown in Figure 7
- [7 marks]** Calculate and draw the Thevenin equivalent of the circuit for terminals a-b
 - [3 marks]** Calculate and draw the Norton equivalent of the circuit for terminals a-b

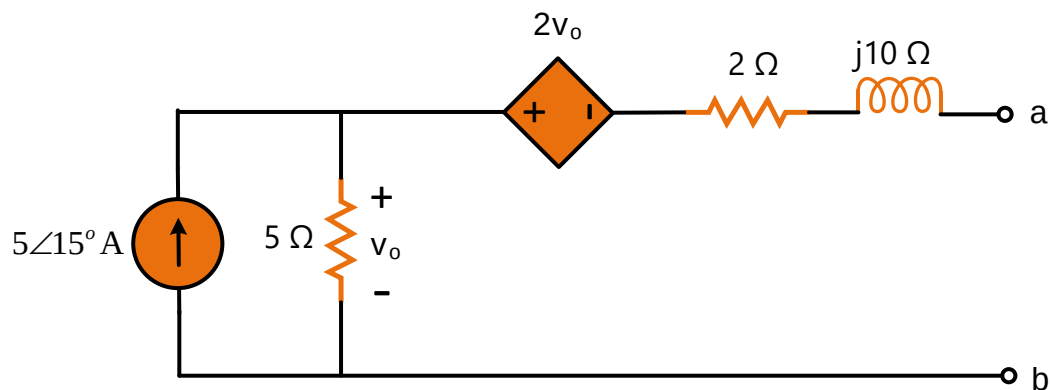


Figure 7

- (iii) **[2marks]** Consider the Thevenin equivalent of a circuit that consists of a voltage source and an impedance Z_{Th} . What is the value of the load impedance Z_L that should be connected to its terminals for maximum power transfer?

IGNORE THIS QUESTION

QUESTION 5 [20 marks]

- (i) [8 marks] Two loads are connected in parallel and are supplied by a $230\text{ V}_{\text{rms}}$, 50 Hz voltage source.

Load 1 consumes 20 kVA of power at a power factor of 0.8 leading (capacitive)

Load 2 consumes 6 kW at a power factor of 0.6 lagging (inductive)

Calculate:

- [2 marks] The total complex power of the two loads.
 - [2 marks] The total apparent power.
 - [2 marks] The combined power factor of the two loads, and
 - [2 marks] The rms value of the current supplied by the source.
- (ii) [6 marks] A ideal 500V:100V transformer is rated at 20 kVA and is supplied by a $500\text{ V}_{\text{rms}}$, 50 Hz voltage source in its primary windings. A load of $10+j10\ \Omega$ is connected in its secondary windings. Draw the circuit and calculate:
- [2 marks] The currents through the primary and secondary windings
 - [2 marks] The power consumed by the load
 - [2 marks] The loading of the transformer (Transformer loading is the ratio of power through the transformer over the rated power of the transformer.)
- (iii) [6 marks] Consider the following logical diagram
- [3 marks] Derive and simplify the logical expression for Z.
 - [3 marks] Write the truth table of the circuit.

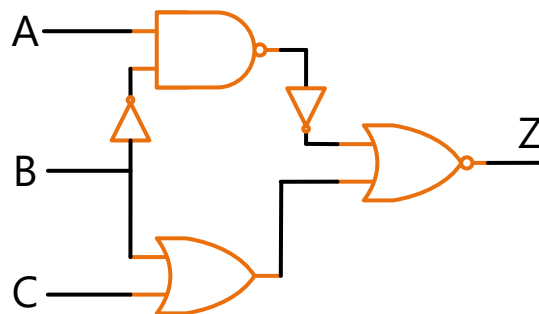


Figure 8

END OF PAPER